

Metropolitan University
Department of Computer Science and Engineering
Spring Term Final Examination - 2026
Program: B.Sc. in CSE; Batch: 58
Course: CSE 417: Software Engineering & Design Pattern

Total Marks: 40

Time: 2 Hours

*[Answer **all** the questions. All parts of each question must be answered sequentially.
Figures in the right margin indicate full marks.]*

(5×2=10)

1. Answer any **FIVE**.

- ☒ (a) Differentiate between White box and Black box testing.
- ☐ (b) If a project has a 'Hard Deadline' and 'Fixed Budget' but the technology stack is completely new, why might an Iterative model be safer than Classical Waterfall?
- ☐ (c) Differentiate between reactive and proactive risk management strategies by providing suitable examples.
- ☒ (d) Write two advantages of Incremental Model.
- ☒ (e) Define Efficiency and Interoperability in terms of requirement engineering.
- ☐ (f) Explain the concept of Backlog Grooming (refinement) and analyze its role in preventing failure during Sprint Planning sessions.
- ☒ (g) List the skills that are required for a software project manager.
- ☒ (h) Define a design pattern and explain its importance in promoting code reusability, and maintainability.

(4×3=12)

2. Answer any **FOUR**.

- ☐ (a) Describe the basic rules, notations, and symbols used in a Data Flow Diagram. Explain the purpose of processes, data stores, and external entities.
- ☒ (b) Differentiate between Alpha Testing and Beta Testing in terms of purpose, environment, and participants. Which one is more effective for obtaining real-user feedback, and why?
- ☒ (c) Your project manager has directed you to prepare a software requirement specification (SRS) for the report. Explain what information should and should not be included in the SRS.
- ☐ (d) Identify the type of risks indicated by the following examples:
 - i. Releasing software that fails in the market.
 - ii. When the project is 70% finished, the client modifies the specifications.
 - iii. Skilled team members exiting the organization.
- ☒ (e) If you were to implement the Agile approach in your web project, what do you consider to be the major benefits and drawbacks of adopting this methodology?
- ☒ (f) Software evolution is a fundamental software activity. In SDLC, evolution for existing systems is addressed in the maintenance phase. What are the general issues that affect a software and how do we handle it?

3. Answer any **THREE**.

(3×6=18)

- (a) A company plans to develop a Smart Healthcare Management System that allows patients to book appointments, access medical records, and receive online consultations. Doctors can update patient records, and administrators manage scheduling and reports. After initial deployment, patients request mobile access and faster response time, while hospitals require integration with existing systems.
- What are the main activities of the requirement engineering process?
 - From the given scenario, identify the stakeholders, the functional and non-functional requirements.
- (b) Write appropriate code for the following scenario:
Design a Notification System that supports multiple delivery methods: EmailNotification, SMSNotification, and PushNotification. Each method sends messages differently. Use the Strategy Design Pattern to enable switching between notification methods at runtime.
Requirements:
- Define an interface NotificationStrategy with a method send(String message).
 - Implement three concrete strategies: EmailNotification, SMSNotification, PushNotification.
 - Create a NotificationContext class to hold and execute the chosen strategy.
 - In the Main class, demonstrate switching between notification strategies.
- (c) Consider FoodExpress, an online food ordering and delivery system. The system allows users to browse restaurant menus, place orders, make online payments, and track delivery status in real time. Using this scenario, draw:
- 0-Level DFD: Identify the system boundary, external entities (e.g., Customer, Restaurant, Payment Gateway), and major data flows.
 - 1-Level DFD: Break down the core processes such as Order Management, and Payment Processing, showing interactions with data stores.
- (d) A national space agency is developing the 'Astra-Link' telemetry module for a satellite launch. The hardware shell is being cast in 6 months, and the software must fit the fixed memory and processor constraints defined today. The internal audit board has a 'Safety-Critical Protocol': For every design requirement, the corresponding system-level test case must be signed off by the Quality Assurance (QA) team before a single line of code is written. While the agency expects the complexity of the signal processing to require multiple internal development cycles to hit performance targets, they have prohibited any experimental code from entering the final integration branch without a pre-existing, verified test-plan mapping.
- Evaluate the scenario and determine appropriate software engineering models.
 - Which models will be best, better, and good for the scenario?
 - Will a hybrid approach or traditional approach be better? Justify your answer with proper explanation.